Investigating Electromagnetic Induction

PhET Faraday's Electromagnetic Lab · PhET

Senior Secondary (Years 11-12)	~30 minutes	Science · Physics

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Simulation: https://phet.colorado.edu/en/simulations/faradays-electromagnetic-lab
Curriculum links: electromagnetism · magnetic fields · electricity

Learning intentions

- I can describe how a changing magnetic field induces a current in a coil.
- I can relate the size of an induced current to the speed of the magnet and the number of coil turns.
- I can explain electromagnetic induction using Faraday's law in qualitative terms.

Getting started

Open the simulation and start on the Pickup Coil (or Bar Magnet) screen. Move the magnet slowly through the coil first and watch the needle on the voltmeter or galvanometer before you try anything faster.

Tasks

1. Move the magnet slowly through the coil. Describe what happens to the needle on the meter as the magnet enters, sits inside, and exits the coil.

2. Now move the magnet through the coil quickly. How does the meter's response differ from the slow-speed trial?

3. Repeat the magnet-through-coil trial at three different speeds and record the peak meter reading each time.

Magnet speed	Peak meter reading

4. Using your table, describe the relationship between the speed of the magnet and the size of the induced current.

5. Increase the number of turns on the coil (if the simulation allows it) and repeat the trial at a constant speed. What happens to the induced current?

6. Explain, in terms of a changing magnetic field, why current is only induced while the magnet is moving relative to the coil, not while it sits stationary inside the coil.

7. If the simulation displays field strength at different points, record the field strength at two different distances from the magnet and describe how field strength changes with distance.

Working:

8. Explain how this simulation demonstrates Faraday's law of electromagnetic induction, and describe one real-world application, such as a generator or induction cooktop, that relies on this principle.

Extension (optional)

Predict what would happen to the induced current if you kept the magnet's speed constant but reversed the direction it moved through the coil partway through a trial. Test your prediction in the simulation and explain the result.